

STA 5635 HW 9

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Problem 1

x_min	y_min	Expresssion 1	Expression 2	Expression 3
0.6828	0.5218	0.1722	0.1722	0.1722

Table 1: Minimized x and y values for the three expressions, as well as the values of the expressions.

Problem 2

Part a

Viterbi sequence:

[illegible]

Part b

The ratio of alpha values at time 118: 0.4970

The ratio of beta values at time 118: 1.3175

Problem 3

$$\hat{\pi} = \begin{bmatrix} 1.0000 & 0.0000 \end{bmatrix}$$

$$\hat{A} = \begin{bmatrix} 0.8698 & 0.1302 \\ 0.2087 & 0.7913 \end{bmatrix}$$

$$\hat{B} = \begin{bmatrix} 0.2128 & 0.2143 & 0.2046 & 0.1896 & 0.0896 & 0.0891 \\ 0.1061 & 0.0921 & 0.1148 & 0.2087 & 0.1861 & 0.2921 \end{bmatrix}$$